

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{4}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_d^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{54} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_d^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_d^{\mathbf{p}}] \right. \\ & \quad \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_e^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_e^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_e^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{36} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \frac{8}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_q^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{108} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_q^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{8}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_q^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_u^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{10}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_u^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \frac{64}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_u^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{27} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{16}{135} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 \text{LF}_{2,1,0}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{2,2,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{36} g_1^4 \text{LF}_{3,1,-1}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{4,1,-2}[m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 \text{LF}_{2,1,0}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{2,2,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{36} g_1^4 \text{LF}_{3,1,-1}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{4,1,-2}[m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{81} g_1^4 \text{LF}_{2,1,0}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{2,2,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{3,1,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{4,1,-2}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{81} g_1^4 \text{LF}_{2,1,0}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{2,2,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{3,1,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{4,1,-2}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{12} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{9} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,1,0}[m_e^{\mathbf{p}}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,2,-1}[m_e^{\mathbf{p}}, \tilde{\mu}] \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{3,1,-1}[m_e^{\mathbf{p}}, \tilde{\mu}] \delta_{i3i4} + \frac{1}{36} g_1^4 \text{LF}_{2,1,0}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 \text{LF}_{3,1,-1}[m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{36} g_1^4 \text{LF}_{2,1,0}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 \text{LF}_{3,1,-1}[m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{24} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \text{LF}_{2,1,0}[m_q^{\mathbf{p}}, \tilde{\mu}] \delta_{i1i2} - \\ & \quad \frac{1}{36} g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \text{LF}_{2,2,-1}[m_q^{\mathbf{p}}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{36} g_1^2 \overline{y}_u^{pi3} y_u^{pi4} \text{LF}_{3,1,-1}[m_q^{\mathbf{p}}, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{2,1,0}[m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{3,1,-1}[m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{2,1,0}[m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{3,1,-1}[m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} +$$